Helmholtz Resonators

Have you ever blown across the top of a wine bottle? If you do it right, you get a clear tone. What pitch would you expect? A wine bottle is about 30 cm long, is closed at one end and open at the other. You might think then you'd have a closed-open pipe, and so the wavelength of the fundamental should be 4 times longer than the pipe, or 1.2m. The speed of sound in air is around 340 m/s, so the fundamenal frequency should be something like $340/1.2 = 283$ Hz. If you actually go and blow on a wine bottle, though, the actual frequency is more like 100 Hz, nearly two octaves lower. Clearly, something else must be going on. Rather than a resonant pipe, the air in the neck of the bottle acts like a mass, and the larger volume of air inside the bottle acts like a spring. Figure 1 shows the basic setup. We have a narrow neck with length L and area A connected to a large chamber with volume V. We'll call the air inside the neck the *plug*, and that plug of air is going to move up and down.

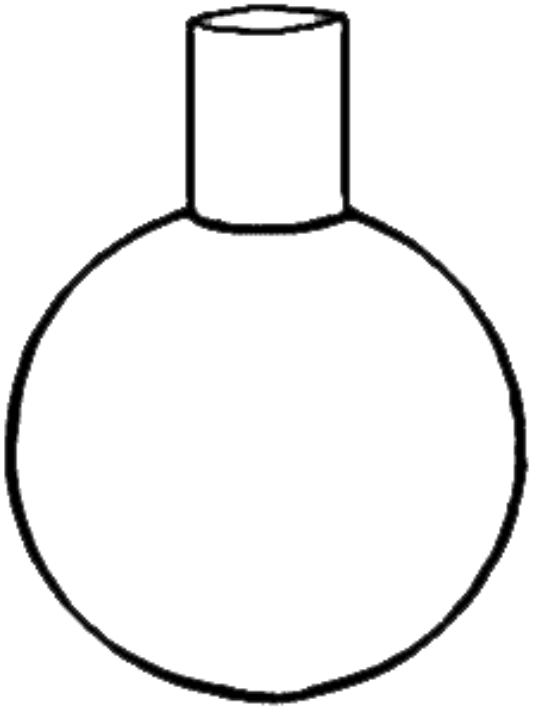


Fig. 1.— Helmholtz resonator. The neck has length L, area A, and the cavity has volume V. The volume of the neck v is the area times length of the neck: $v = AL$.

1. Helmholtz Frequency

Now that we have basic setup, how will the system behave if we jostle the air in the neck? The picture you should have is that the air in the neck acts like a solid mass and it's going to move up and down. As it does, it compresses/expands the air inside the bottle and so there's going to be a

pressure. To a very good approximation, the pressure where the neck meets the atmosphere stays at the background atmospheric pressure, so the only force on the air in the neck will be the air in the chamber pushing back.

What happens if we move the air in the neck down by a distance x? That mean we're squeezing a volume Ax out of the way, so if the air in the chamber started out with volume V, now that same air is in a volume $V - Ax$. The volume went down by a factor of $V/(V - Ax)$ so the density had to go up by a factor of:

$$\rho_{new} = \rho_0 \frac{V + Ax}{V} = \rho_0 \left(1 + \frac{Ax}{V}\right) \tag{1}$$

where ρ_0 is the background density of the air and ρ_{new} is the density after we pushed our air plug down. The *change* in density $\delta\rho \equiv \rho_{new} - \rho_0$ is then:

$$\delta\rho = \rho_0 \frac{Ax}{V} \tag{2}$$

Equation 2 tells us the change in density, but if we want to know the force on the plug, we have to work out the change in *pressure*. Normally this can be quite difficult to do, but if you remember back to the note on sound air, you may recall that the speed of sound c_s is:

$$c_s^2 = \frac{dP}{d\rho} \tag{3}$$

which I can re-write as:

$$dP = c_s^2 d\rho \tag{4}$$

We can plug Equation 2 into Equation 4 to get

$$dP = c_s^2 \rho_0 \frac{Ax}{V} \tag{5}$$

Equation **??** tells us the *difference* in pressure between the top and bottom of the plug, and so the net force on the plug is the area times the pressure, or:

$$F = AdP = \frac{c_s^2 \rho_0 A^2 x}{V} \tag{6}$$

Equation 6 gives us the force, but what we really want is the acceleration. As usual we'll use Newton's second law. The mass that moves is the mass of the air in the plug. The volume of air there is the area times the length or AL, and the mass is the density times the volume, or $m = \rho_0 AL$. We now have the acceleration is:

$$a = \frac{F}{m} = \frac{c_s^2 \rho_0 A^2 x}{\rho_0 V AL} = \frac{c_s^2 Ax}{VL} \tag{7}$$

Which way is the force/acceleration? Well if x is positive then the volume in the chamber went up, so its pressure went down. That means the acceleration will be *opposite* to the change in position x. Since the acceleration is the rate of change of velocity and velocity is the rate of change of position, the acceleration is the rate of change of the rate of change of position. We can write this down as an equation:

$$\frac{d^2x}{dt^2} = -\frac{c_s^2 A}{VL} x \tag{8}$$

This is what is called a *differential equation*, and actually solving them is beyond the scope of this course. In words, though, we're looking for how the height x of the plug depends on time, and we know that the slope of the slope of the height is equal to a (negative) constant times the height.

I'll give you two facts from calculus: the slope of $\sin(t)$ is $\cos(t)$, and the slope of $\cos(t)$ is $-\sin(t)$. Recall that if we want the slope of $\sin(\omega t)$, it's ω times the slope of sin, so the slope of the slope (the second derivative) of $\sin(\omega t)$ is:

$$\frac{d^2(sin(\omega t))}{dt^2} = -\omega^2 \sin(\omega t) \tag{9}$$

If we compare Equation 9 to Equation 8 then we see they agree, as long as

$$\omega^2 = \frac{c_s^2 A}{VL} \tag{10}$$

or

$$\omega = \sqrt{\frac{c_s^2 A}{VL}} \tag{11}$$

This is what is known as *simple harmonic motion* and is ubiquitous in physics and the world around us. More generally, if I have an object, and the force on the object is proportional to how far I moved the object but in the opposite direction (so the force wants to *restore* the object to its original positio), then if I whack the object, its position is going to be a sine wave. The frequency of the oscillating position is set by (the square root of) the restoring force. Notice that the frequency of the oscillations does *not* depend on how *big* the oscillations are. If I pull my object further away from its resting point, it has to move further to get back to where it started, but the force also went up by exactly the right amount so the *time* it takes to get back remains the same. A system that behaves this way is called a *simple harmonic oscillator*.

Equation 11 tells us that the height of our plug will be proportional to $\sin(\omega t)$. The *period* of the oscillation will be when $t = 2\pi\omega$ and the *frequency* will be the usual $\nu = \omega/2\pi$. We can plug that in to get the frequency of our Helmholtz resonator:

$$\nu = \frac{1}{2\pi}\sqrt{\frac{c_s^2 A}{VL}} \tag{12}$$

This is what you'll see if you look in *e.g.* Wikipedia, but there's a form I think is more useful. If we multiply the top and bottom of the square root insides by L, we get

$$\nu = \frac{1}{2\pi}\sqrt{\frac{c_s^2 AL}{VL^2}} \tag{13}$$

We know the volume of the plug is $v = AL$, and anything that shows up *squared* we can pull out of the square root, which leaves:

$$\nu = \frac{c_s}{2\pi L}\sqrt{\frac{v}{V}} \tag{14}$$

Equation 14 tells us intuitively what's going on. The first factor $\frac{c_s}{2\pi L}$ is the frequency of an oscillation with wavelength equal to the length of the neck. Note that it is a full wavelength, not a half or quarter. The air in the cavity is pretty squishy, though, so that natural frequency is going to get reduced. The reduction factor is the square root of the ratio of volumes. The bigger the chamber, the lower the pressure is when the plug moves, and so the lower the frequency.

2. Resonators in Practice

Given how simple the physics was that we just looked at, you might expect Helmholtz resonators to be everywhere. You would be correct. Whenever we have a hole connected to a chamber, we'll get a resonator. Wine bottles are one example, but there are many, many others. If you've ever driven a car with one window cracked, you may have noticed a very low-frequency thumping. That's a Helmholtz resonance, with the air in the window acting as a plug and the car acting as a cavity. There are several instruments that are Helmholtz resonators. Perhaps the best-known one is the ocarina (at least, best-known amongst Nintendo afficianados). The ocarina has a hollow chamber with several holes. As you cover/uncover holes, you are changing the mass of air that is going to oscillate, and ocarina will change pitch accordingly.

Less obvious but also true is that *body* of a stringed instrument is a Helmholtz resonator. The {-hole acts like a neck, and the air inside the instrument is the resonating mass. The string itself hardly moves any air, but it pushes on the bridge and the bridge in turn shakes the body in a way that resonates. That *amplifies*[1] the sound from the vibrating strings. If you removed the body, a violin would be almost inaudible. A violin is tuned so that its Helmholtz resonance is at a concert D matching the second string of the violin, which is why so many violin concerti are in D-major/minor.

Helmotz resonators have tons of other practical applications. One common speaker design is *vented* or *ported* speakers, which have an extra hole in the cabinet, maybe with a tube. That tube acts as a Helmholtz resonator, and really helps the speaker produce more bass in a smaller volume. Historically, it was really hard to design a speaker where this sounded good, because if you have a sharp resonance then one note would be really loud but others would be much softer. Engineers figured this out in the '60's and '70's, and now these designs are ubiquitous. The Helmholtz resonance is fairly wide, and it's designed to interact with other resonances in the speaker system to give a broad, flat response as a function of frequency.

Helmholtz resonators are also widely used in reverse to convert sound into mechanical energy. Why would you want to do this? Well, it's easy to damp mechanical energy, so if I have a loud sound that I want to get rid of, I can use a Helmholtz resonator to suck up that energy and dissipate it as heat. Usually you do this by putting in some lossy filler in the large cavity that damps out the vibrations. HVAC systems and airplanes commonly do this. Concert halls do as well - rooms naturally have frequencies that they resonate at, just like pipes, and you *don't* want your concert hall to be louder for some notes than others. Engineers put in panels with absorptive Helmholtz resonators tuned to damp down the room harmonics to flatten the response of the hall. There's a nice writeup here of the refurbishment of a concert hall that covers the intial design (with pictures!) and restoration of the hall's Helmholtz resonators.

[1] I am using amplify loosely here. Technically, amplify to me means you add energy to a system, but that's not what's going on. Instead, the job of the body of the violin is to do a more efficient job of turning the vibrations of the string into vibrations in air. It doesn't add energy, but it does greatly speed up the rate at which the vibrational energy turns into sound energy.

3. Worked Example - Wine Bottle

Let's work out how a wine bottle behaves and see if we can predict the frequency at which it will resonate. Measuring volumes of irregular containers can be hard, but we can pour water into them and measure the change in weight. Since water is very close to 1 gram per cubic cm, we can translate a change in weight to a volume.

I took a standard bottle of wine and a kitchen scale, and zeroed the scale with the bottle empty. Then I poured water into the bottle up to the neck and found it weighed 734 grams more. I finished filling the bottle, which took another 29 grams of water. My best guess for the length of the neck was 7 cm. The measurement is tricky, though, because in practice the air plug can extend past what looks like the neck (*e.g.* some of the air just above the bottle also vibrates, adding to the effective mass). If we take those numbers at face value, the predicted frequency is

$$\frac{340}{2\pi 0.07}\sqrt{\frac{29}{734}} = 153\ Hz \tag{15}$$

However, when I go and actually play the bottle, the frequency is noticeably lower, about 108 Hz. Figure 2 shows the loudness of the bottle vs. frequency, with the fundamental and its overtones marked by red dots. The first two overtones come through clearly, but there are no clear overtones past those.

The data in Figure 2 actually let us do a much better job of predicting the Helmholtz frequency. The neck of the bottle can also act as resonant pipe that is open on both ends. If we could ignore the Helmholtz noise, we'd hear a high-pitched whistling from the neck. That in turn tells us how long vibrating column of air in the neck is, and is a lot more accurate than we're likely to be with a ruler. The first big bump after the red dots is that mode, and happens at around 1630 Hz. If we take 340 m/s as the speed of sound, and remember that an open-open pipe is half a wavelength, then we have

$$2L = \frac{340}{1630} \tag{16}$$

$$L = .104m \tag{17}$$

If we use *that* length and plug into Equation 15, we get

$$\frac{340}{2\pi 0.104}\sqrt{\frac{29}{734}} = 103\ Hz \tag{18}$$

Now that is *very* close to the measured value of 108 Hz. Since we used a shorter neck than we should have when estimating the neck length, we probably also underestimated the volume of the neck, which would drive the frequency higher. Resonators are everywhere, and with some cleverness, we can predict their oscillation frequencies!

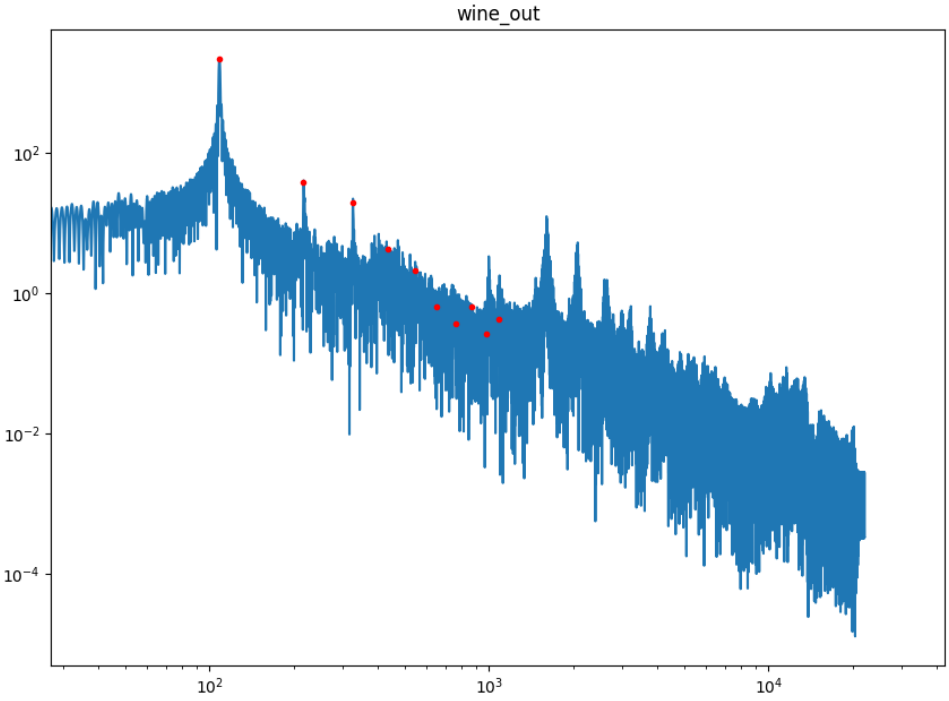


Fig. 2.— Frequency spectrum of wine bottle. The fundamental is 108 Hz is marked by the first red dot, and the other red dots are at integer multiples of that. You can see two overtones clearly, but nothing past that. The largest bump after that at around 1630 Hz comes from the neck acting as an open-open pipe resonator. We can use that length to measure the effective length of the neck.